TABLE OF CONTENTS

SECTION 1-INDIGO FOSA PROGRAM

DEFINITION OF FOSA: 4

WHY FOSA: 5

FOSA OPERATING CHARACTERISTICS: 6

SECTION 2-OBSERVER ROLES AND RESPONSIBILITIES

FOSA OBSERVER: 10

RATIONALE AND ETIQUETTE: 10

FOSA DATA QUALITY: 11

A DAY IN THE LIFE OF AN OBSERVER: 12

NARRATIVE WRITING: 13

SECTION 3-THREAT ERRROR UAS

THREAT AND ERROR (TEM) MANAGEMENT: 16

INTRODUCTION: 16

WHAT IS TEM? 16

THREAT AND ERROR MANAGEMENT: 16

THREAT MANAGEMENT 17

DEFINITION 17

TYPES OF THREATS: 17

THREATS RESPONSE: 18

SUMMARY OF THREAT MANAGEMENT: 18

ERROR MANAGEMENT 19

DEFINITION: 19

TYPES OF ERRORS 20

ERROR RESPONSE: 20

ERROR MANAGEMENT: 21

UAS MANAGEMENT: 22

DEFINITION: 22

MANAGEMENT OF UAS: 22

SUMMARY OF UAS MANAGEMENT: 23

TEM SUMMARY 24

COMPETENCY: 24

COMPETENCIES: STATE OF THE ART TEM COUNTERMEASURES: 24
"MONITORING" AS A COUNTERMEASURE:.....26
INDIGO FOSA EXCEL SHEET:.....27
GETTING STARTED:.....28
DEMOGRAPHIC WORKSHEET:.....29
THREAT MANAGEMENT WORKSHEET:.....31
ERROR MANAGEMENT WORKSHEET: 35
UAS MANAGEMENT WORKSHEET: 42
SUBMITTING THE EXCEL SHEET: 45

ANNEXURE 1-SAMPLE NARRATIVES
GOOD NARRATIVE EXAMPLES: 48
POOR NARRATIVE EXAMPLES: 52
ANNEXURE 2 - OBSERVER BIAS
OBSERVER BIAS 52
ANNEXURE 3-TEM CODE BOOK
THREATS.....58
ERROR 60
UAS.....66
ANNEXURE 4-COMPETENCY CARD
COMPETENCY CARD 54
ANNEXURE 5-FOSA TEAMS CHANNEL
FOSA TEAMS CHANNEL.....56
ANNEXURE 6-IMPORTANT CONTACTS
IMPORTANT CONTACTS.....59

SECTION I:
INDIGO FOSA PROGRAM

DEFINITION OF FOSA:

A FOSA (an in-house LOSA program) is a formal process that requires expert and highly trained observers to ride the jump-seat during regularly scheduled flights to collect safety-related data on environmental conditions, operational complexity, and flight crew performance. Confidential data collection and non-jeopardy assurance for pilots are fundamental to the process.

The FOSA program at IndiGo incorporates all the operating characteristics of the Line Operations Safety Audit program detailed in ICAO doc 9803 and FAA AC 120-90.

Using a medical metaphor, a FOSA is similar to a patient's annual physical examination. People have comprehensive checkups in the hope of detecting serious health issues before they become consequential. A set of diagnostic measures, such as blood pressure, cholesterol, and liver function, flag potential health concerns, which in turn suggest needed changes to the patient's current lifestyle. *A LOSA is built upon the same proactive notion. It provides a diagnostic snapshot of strengths and weaknesses that an airline can use to bolster the "health" of its safety margins and prevent degradation.*

FOSA is distinct from - but complementary to - other predictive safety programs such as electronic data acquisition systems (e.g. FOQA, voluntary reporting systems, etc)
However, these programs have three major conceptual differences.

Differences between a FOSA program and other safety programs:

1) First, FOQA and voluntary reports rely on outcomes to generate data. For FOQA, it is flight parameter exceedances; for Voluntary Reports, it is adverse events that crews report. By contrast, FOSA samples all activities in normal operations. In these regularly scheduled flights, there may be some reportable events, but there will also be some near-events and, importantly, a majority of well-managed, successful flights. FOSA provides a unique opportunity to study the flight management process, both successful and unsuccessful, by noting the problems crews encounter on the line and how they manage them.

2) The second major difference is the perspective taken by each program. With its focus on electronic data acquisition downloaded directly from the aircraft, FOQA can be said to have the "airplane perspective." Voluntary reporting provides the "pilot perspective" by using pilots' voluntary disclosure and self-reporting of events. Voluntary reports provide insight into why events occur as seen from the crew's perspective. By contrast, FOSA provides a "neutral, third-party perspective" in that FOSA observers record contextual and flightcrew data on every phase of flight, regardless of the outcome. All three perspectives provide useful data to an airline's safety management system.

3) The third major difference is that FOSA unlike the FOQA, Voluntary reporting and Surveillance audit flights is that it is completely anonymous and non-punitive. FOSA is not a compliance audit.

What it is…

FOSAs are non-jeopardy, peer to peer, observational flights that employ a standardize methodology to provide managers and air crews with information in managing operational risk..

FOSA detects procedural "drift" in operational practices, perhaps towards less safe practices, or, by adaptation towards safer practices and outcomes.

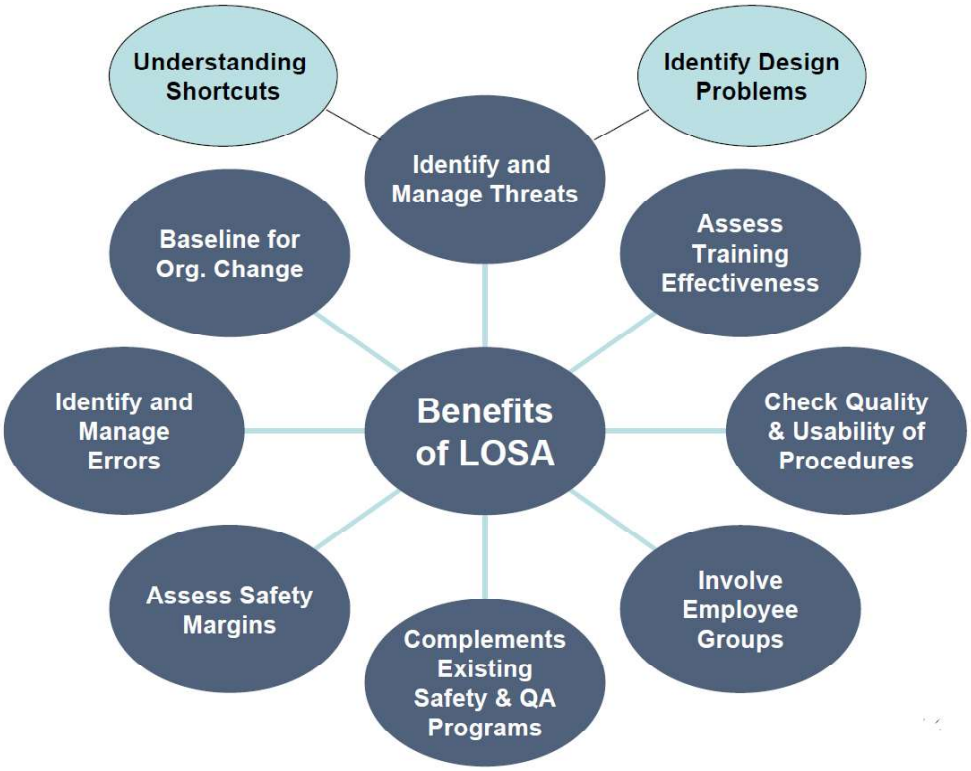
FOSA looks at organizational resilience through the ability of aircrews to manage both their own errors and disturbances to the system.

What it isn't…

FOSA is NOT a compliance audit, it is a data source that provides "Lead Indicators: for mishap prevention purposes.

FOSA IS NOT A CHECK RIDE

WHY FOSA:



FOSA OPERATING CHARACTERISTICS:

FOSA is defined by the following ten operating characteristics that act to ensure the integrity of the FOSA methodology and its data. Without these characteristics, it is not a FOSA. These characteristics are:

1. **Jump-seat observations during normal flight operations:**

   FOSA observations are limited to regularly scheduled flights. Line checks, initial line indoctrination or other training flights are off-limits due to the extra level of stress put upon the pilots during these types of situations. Having another observer on board only adds to an already high stress level, thus providing an unrealistic picture of performance. In order for the data to be representative of normal operations, FOSA observations are collected on regular line flights.

2. **Joint management / pilot sponsorship**:

   In order for FOSA to succeed as a viable safety programme, it is essential that both leadership and pilots (through their professional association, if it exists) support the project. The joint sponsorship provides a "check and balance" for the project to ensure that change, as necessary, will be made as a result of FOSA data.

3. **Voluntary crew participation:**

   Maintaining the integrity of FOSA within an airline and the industry as a whole is extremely important for long-term success. One way to accomplish this goal is to collect all observations with voluntary crew participation. Before conducting FOSA observations, an observer must first obtain the flight crew's permission to be observed. The crew has the option to decline, with no questions asked. The observer simply approaches another flight crew on another flight and asks for their permission to be observed.

4. **De-identified, confidential and safety-minded data collection:**

   FOSA observers are not to record names, flight numbers, dates or any other information that can identify a crew. This allows for a level of protection against disciplinary actions. The purpose of FOSA is to collect safety data, not to punish pilots. Pilots must not be fearful that a FOSA observation could be used against them for disciplinary reasons. ***If a FOSA observation is ever used for disciplinary reasons, the acceptance of FOSA within the airline will most probably be lost forever***.

5. **Targeted observation instrument**:

   The current data collection tool to conduct a FOSA is the FOSA Observation Form based on Threat and Error Management (TEM). Data collection instrument used needs to target issues that affect flight crew performance in normal operations. Flight and crew demographics such as city pairs, aircraft type, flight time, years of experience within the airline, years of experience within position, and crew familiarity;

   - Written narratives describing what the crew did well, what they did poorly and how they managed threats or errors for each phase of the flight;

- Competency ratings using research developed observable behavioural markers for each competency observed;
- Threat management worksheet that details each threat and how it was handled;
- Error management and UAS management worksheet that lists each error/UAS observed, how each error was handled and the final outcome; and
- Crew interview conducted during low workload periods of the flight, such as cruise, that asks pilots for their suggestions to improve safety, training, and flight operations.

6. **Trusted, trained and calibrated observers**:

   Primarily, pilots conduct FOSAs. Observation teams will typically include line pilots. It is critical to that observers are respected and trusted within the airline to ensure the line's acceptance of FOSA. Good observers are the lifeline of FOSA. **If an observer is unmotivated or untrustworthy, FOSA will fail**. All observers are selected, everyone is trained and calibrated in the FOSA methodology, including the use of the FOSA forms and, particularly, the concepts of threat and error management. Training of observers in the concepts and methodology of FOSA ensures that observations will be conducted in the most standardized manner. After completing training, observers observe regularly scheduled line flights. The objective is to observe the largest number of crews and segments possible in the time frame, given the flight schedules, logistics and types of operation sampled.

7. **Trusted data collection site**:

   In order to maintain confidentiality, IndiGo has a trusted data collection site on the MS platform. This ensures that no individual observations will be misplaced or improperly disseminated through the airline.

8. **Data verification**:

   Data-driven programmes like FOSA require quality data management procedures and consistency checks. For FOSA, these checks are done at by trained 'validators' who scan the raw data for inaccuracies. For example, an observer might log a procedural error for failure to make an approach callout for which there are actually no written procedures in the airline's flight operations manual. Therefore, it would be the job of the validating team to detect and delete this particular "error" from the database. The end product is a database that is validated for consistency and accuracy according to the airline's standards and manuals, before any statistical analysis is performed.

9. **Data-derived targets for enhancement:**

   The final product of a FOSA is the data-derived FOSA **targets for enhancement**. As the data are collected and analysed, patterns emerge. Certain errors occur more frequently than others, certain airports or events emerge as more problematic than others, certain SOPs are routinely ignored or modified and certain manoeuvres pose greater difficulty in adherence than others. These patterns are identified for the airline as FOSA targets for enhancement. It is then up to the airline to develop an action plan based on these targets, using experts from within the airline to analyse the targets and implement appropriate change strategies.

10. Feedback of results to the line pilots:

After a FOSA is completed, the airline's management team and pilots' association have an obligation to communicate FOSA results to the line pilots. Pilots will want to see not only the results but also management's plan for improvement. If results are fed back in an appropriate fashion, experience has shown that future FOSA implementations are welcomed by pilots and thus more successful.

SECTION II:
OBSERVER ROLES & RESPONSIBILITIES:

FOSA OBSERVER:

Rationale and Etiquette:

Observers will likely have a rudimentary understanding of FOSA when selected for the project; however, they will need to fully understand the safety rationale for conducting a FOSA at their airline. A "big picture" perspective will help observers understand the "why" of FOSA and will understand the importance of their role in the FOSA process. Also, the observers will be ambassadors for FOSA while observing on the line, and it is important that they be able to explain the process fully, to answer any questions that the line pilots may have, and allay any fears or concerns.

An observer needs to learn how to approach a crew, how to ask permission to observe a flight, how to walk away so that the crew can discuss it, and to accept without question any crew's decision to deny access to the cockpit. The observer should also carry a copy of the PIN signed by management to show any interested crew.

Where to meet the crew?

- ***FOSA observation NOTE is reflected on the operating Flight Crew's roster. Hence, best practice is to meet in Dispatch before the crew arrives (get there first)***

On the jumpseat, the observer's behaviour is best summarized as **"a fly on the wall."** The observer needs to be unobtrusive, yet responsive to any queries the crew may have. The observer does not complete the Narrative on the jumpseat—this would be a disconcerting distraction to the crew. Instead, the observer is encouraged to carry a small pocket notebook to jot down minimalist notes, just enough to jolt the memory when outside the cockpit and completing the full Narrative.

"Fly on the wall" is an observational technique that allows an Observer to collect data by seeing and listening. Usually, observer employ this method to gain insight into people, environment, interactions and objects in a space. It is the primary responsibility of the observer to stay completely unnoticed during the observation so as to not bias the participants in any way.

- **Be aware of making noise (no pens that click)**
- **Do not fill the cockpit with light when taking notes**

FOSA observers should accept their role as observers, not evaluators—they are not 'check' pilots. FOSA observers will observe errors and undesired aircraft states as part of their observations; however, they should only interrupt if they perceive the safety of the flight to be seriously and immediately endangered. ***A helpful rule of thumb is to think of themselves as a guest on the jump seat of another airline. This***

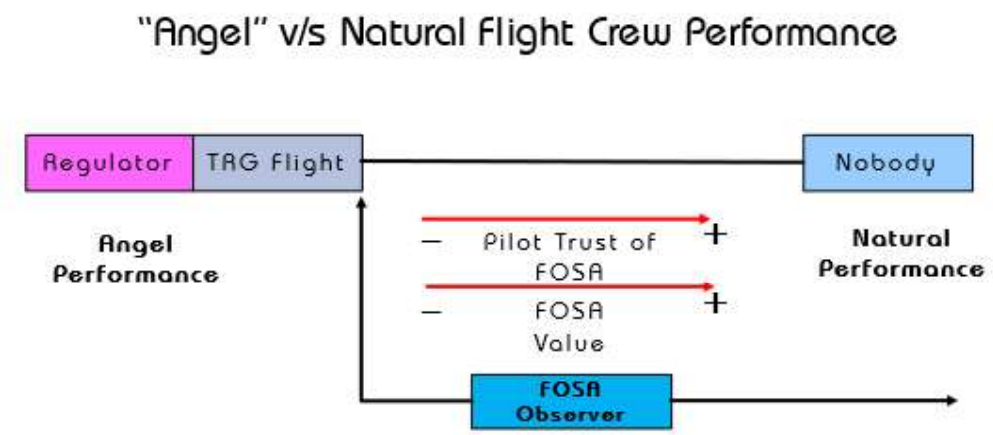
seems to help distance the observer from the flight crew while still being able to politely point out safety concerns if they come about.

The FOSA observer is the only person in the whole FOSA process who has access to crew identities. It is essential, therefore, that observers are keep in mind that confidentiality is critical. Confidentiality is paramount and observed crew behaviour should not be discussed with anyone—not even other observers. Experience has shown that at the end of a flight, crews will often ask the observer to "debrief" their performance. In these circumstances, it is essential that the observer politely decline the invitation. This emphasizes the concept that the observer is not there to evaluate the crew, merely to record events.

FOSA observers should act in an unobtrusive and consistent manner so that line pilots have a similarly positive experience of FOSA, which in turn will favourably affect their receptivity to the final results and outcomes.

FOSA Data Quality:

In order to collect high quality data from a FOSA observations it's imperative to really see what happens on the line, there has to be a guarantee of confidentiality with no regulatory or organizational jeopardy for the crews that were observed. Crews have to believe there would be no individual repercussions; otherwise, they would revert to their best "angel performance" when being observed and the audit would uncover nothing more than what was learned from line check or training data.



Objective for FOSA is to get as close as possible to collecting natural flight crew performance data

HIGH PILOT TRUST = HIGH QUALITY FOSA DATA

A Day in the Life of an Observer:

Typically, the FOSA flight would be reflected on the FOSA Observers roster. Roster will be released with 2 OFDs planned. Post roster release the first OFD will change to an observation flight with an appropriate roster code. Flights that are planned will be updated on an excel sheet at the back end. In case of a particular observation could not to be undertaken, the back end team needs to be informed.

1) Pre-flight BA is a requirement for the observation.

2) On reaching the dispatch, the observer will first get the flight crew's permission to conduct a FOSA Observation on the said flight. Observer may use the company NOTAC or SVP FLIGHT OPS email to clarify any questions that the flight crew have regarding the FOSA flight.

3) Observer must prepare for the flight just as they would if they were operating.

4) Observers IGA will have to be entered in the ATSU INIT page as a SNY and in the Tech Log as an observer.

5) Throughout the flight, the Observer will follow the "Fly On The Wall" approach and take notes discreetly to avoid distracting the flight crew.

6) On completion of the flight, the observer will be allotted one OFD (office day) to complete the report.

7) On completion of the flight the Observer will write a narrative of the flight and follow the steps detailed in CODING:IndiGo FOSA excel sheet

8) On completion of the report, the Observer will log on to the team's channel to update their report status. (Refer to Annexure V)

SUMMARY:

Observer to Crew Briefing

1. Introduce yourself as a FOSA observer
2. Show FOSA PIN //email
3. Stress you are only there to collect safety data – all are kept confidential
4. Obtain the crew's permission to observe
5. If the crew declines, thank them for their consideration and walk away
6. Once permission is granted:
   i. Schedule a crew interview time
   ii. Inform the cabin crew with Captain permission
   iii. Turn into a "fly on the wall'

Narrative Writing:

Narrative writing is a crucial part of filling the FOSA report. FOSA Observers are the eyes and ears for the rest of the Indigo FOSA team (Data Validation and Data Analysis team). They form the base for all the further reports and hence a high-quality narrative is imperative to maintain a good standard for the FOSA report.

A narrative written by the observer should have sufficient detail to allow others to understand the flight and all its events. The observers need to describe the environmental conditions and events surrounding the pilots' behaviour so that the crews' performance can be understood in full context.

The observers should realize that a good-quality narrative is imperative. If need be the observer is encouraged to "overwrite" the flight rather than provide too little detail. In particular, observers need to record events that happened, such as threats, errors, or undesired aircraft states, the context in which they happened, and the crew's response and management of the event.

Observers are selected because they are experts at understanding flight operations and this expertise is best expressed in detailed narratives. As long as the observer provides a detailed narrative of the flight, any coding oversights can be remedied later in the data-validation process - ***a good-quality narrative is the ultimate key to standardized data.***

TIPS TO WRITE A GOOD NARRATIVE:

- A narrative must cover two aspects:
  - The Crew performance: A description of how the crew managed their Threats Errors and UAS.
  - The Operational context: Is the continual factors that shapes the crew performance, builds the storyline and fills the gaps for anyone who reads report for back-end validation purpose.

The right balance of crew performance and operational context for a good narrative would be:

75-80% - CREW PERFORMANCE
20-25%- OPERATIONAL CONTEXT.

- Provide data that no other data source can provide. Remember that the observer is the only data source that can provide the company of information on HOW a threat has been managed and WHY an error has occurred.

- Justify blanket statements: Don't just write "the crew did a good job"
  Need to justify why crew did a good job
  Remember – we know 'what' through other sources of data. We need to know 'why' …. In NORMAL OPERATIONS.

- Keep your narrative and observation Non-Judgemental. Justify your narrative with factual observations.

For Example: Observation of PIC Leadership:
The captain did an unremarkable job of leading an experienced and competent FO. While he picked up a couple of important omissions from briefings, he was too confident in his FO's ability and subsequently failed to detect the altitude setting error during the approach (Marginal Leadership).

For some examples of good/poor narratives refer to Annexure I

- The Narrative should be divided phase wise with the following phases:
  - **INTRODUCTION:** The introduction should give a context for the entire flight.

    Example: Flight sector, Base/Transit flight, Task allocation for the sector (PF/PM), any Internal threats observed (Fatigue, newly released Capt/FO, etc)

  - **PRE DEPARTURE/TAXI OUT**
  - **CRUISE**
  - **DESCENT/APPROACH/LANDING/TAXI IN**

    For the above phases of flight the narrative should provide context. What did the crew do well? What did the crew do poorly? How did the crew perform when confronted with Threats, Crew Errors and UAS.

  - **OVERALL FLIGHT:**
    - The 'Overall flight' should bring out the Team climate for the entire flight i.e. the narrative should provide an overview of Leadership and Communication attributes displayed by the flight crew for the entire flight.
    - No new threats, errors or UAS should be written in the 'Overall flight'.
    - The 'Overall flight' narrative must highlight the overall conduct of the flight specifically highlighting the LTW and COM competencies (the tone of the flight, assertiveness and capability of speaking up (FO), understanding each other's roles and responsibilities etc.
    - The 'Overall flight' narrative must not be a repeat of what is already written in the earlier flight phases.

- The Final Narrative should be highlighted using the following colour codes:
  - Green – Threats.
  - Yellow – Errors.
  - Red – UAS.
- The Word document must be renamed to the UIN number and the same should be uploaded on to the individuals teams channel.

SECTION III:
THREAT ERROR UAS

THREAT and ERROR (TEM) Management:

Introduction:

The Threat and Error Management (TEM) model is a conceptual framework for understandings operational performance in complex environments. Originally created to capture the flight crew's task in commercial aviation, the model is generic and can be applied to numerous work situations. The added value that TEM brings to other performance models is that it focuses simultaneously on the operating environment and the humans working in that environment. Because the model captures ongoing performance in its "natural" or normal operating context, the resulting description is realistic, dynamic, and holistic. Because the model can also quantify the specifics of the environment and the effectiveness of performance in that environment, it is also highly diagnostic.

There are several ways of using the TEM model, from focusing on a single event (as is the case with accident/incident analysis) to understanding systemic patterns in a large set of events (as with Line Operations Safety Audits (LOSA)). As a training tool, TEM can help individuals clarify their performance needs and vulnerabilities; and as part of a safety management system, TEM can help an organization measure and improve the effectiveness of its organizational
defences and safeguards.

"FOSA is grounded within the threat and error management (TEM) framework"

What is TEM?

TEM provides a way for pilots to look for potential threats to fight operations in a structured way. They actively manage these threats and any errors that may lead to undesired aircraft states and therefore to the safety of the fight. TEM encompasses training, briefings, checklists, standard operating procedures, and human factors principles.

Threat and Error Management:

- Recognise and manage errors
- Recognise and manage threats
- Recognise and manage undesired aircraft states.

TEM involves:

- planning to identify threats and errors during a fight and implementing countermeasures to eliminate or minimise them
- directing actions to address threats and errors, using checklists, approved procedures, and other acceptable means, including self-direction during single-pilot operations
- controlling the progress of events to ensure a safe outcome. This step requires monitoring progress and amending plans and actions as required, including correcting any undesired aircraft state (UAS).

THREAT MANAGEMENT

Definition

External events or situations that:

- increases the operational complexity of a flight
- Occur outside the influence of the flight crew
- Requires immediate crew attention to maintain safety margin

Threat is a situation or event that has the potential to have a negative effect on fight safety, or any influence that promotes an opportunity for pilot error/s.

Threats are generally external (such as bad weather) or internal (such as physiological and psychological state).

Types of Threats:

There are two types of threats.

INTERNAL THREATS:

Internal threats are related to the individual, however these threats are outside the influence of the flight crew and need to be mitigated.

Examples of internal threats are:

- fatigue
- experience
- lack of recency and proficiency

EXTERNAL THREATS:

.

External threats can relate to the context of the operation and therefore can be different depending on the situation.

Examples of external threats are:

- adverse weather
- high terrain or obstacles (wire, etc.)
- night operations
- other traffic
- equipment faults
- unanticipated turbulence
- unruly passengers
- smoke in cabin
- equipment faults
- Interruption/distraction

Threats Response:

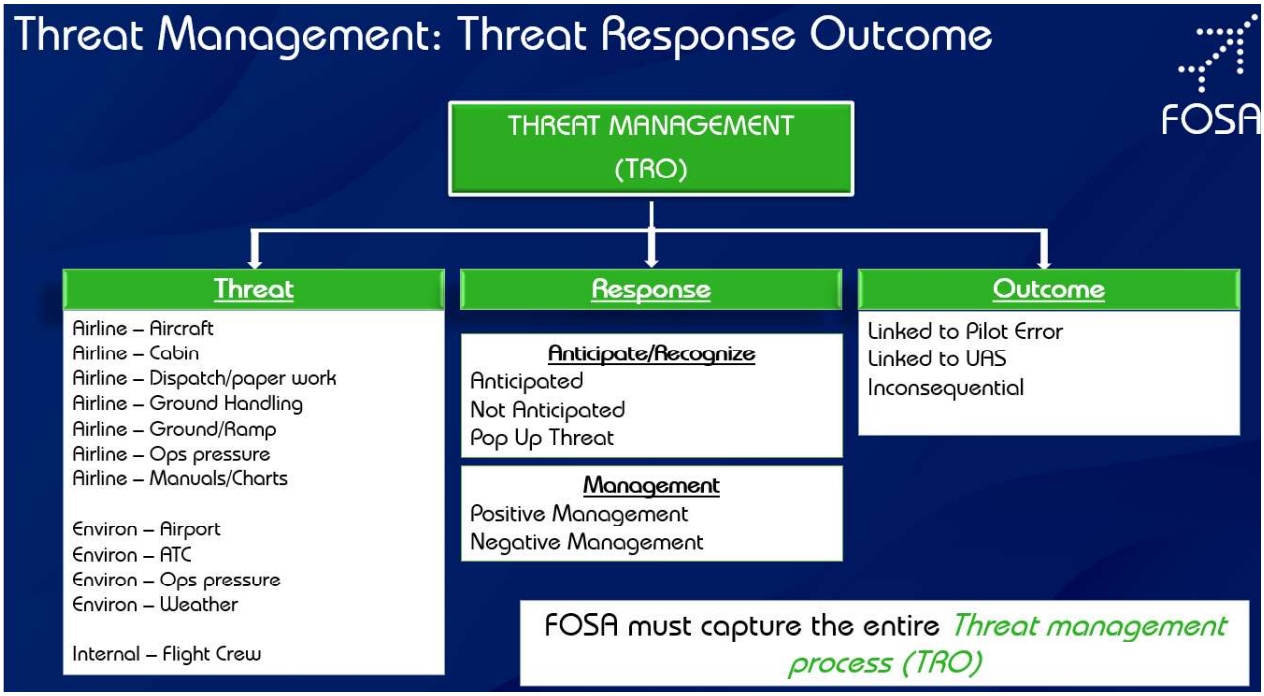
The TEM model includes three threat management categories: *anticipated*, *unanticipated* and *latent*. All three can reduce safety margins.
Latent threats may not be clear and may need to be uncovered by formal safety analysis and specifically addressed in organization's training and procedures.

1) **Anticipated:** Detected / discussed ahead of time
   E.g. Crew identified the possibility of a runway change after pushback and discussed the mitigation actions

2) **Not Anticipated:** Not discussed ahead of time when it could have been
   E.g. Crew did not discuss how they were going to manage adverse weather conditions during descent/ approach

3) **Pop Up Threat:** Anticipation not possible
   E.g. During taxi – out, a malfunction occurred or a ATC issued a change in runway/clearance.

Summary of Threat Management:



ERROR MANAGEMENT

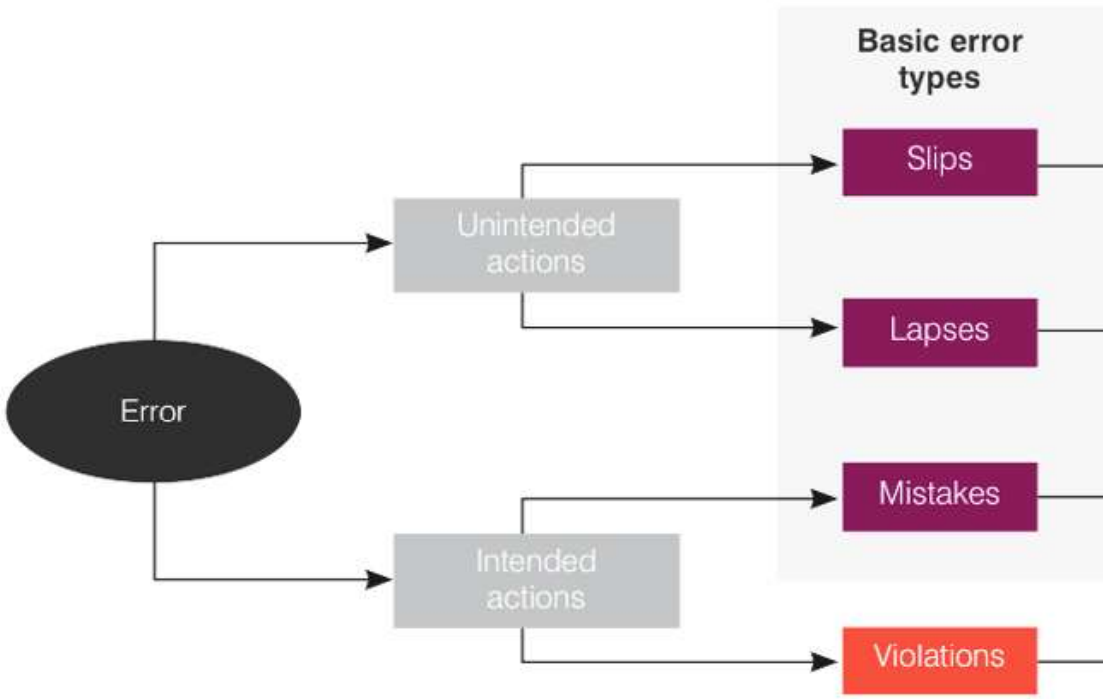
Definition:

Errors are defined as flight crew action or inaction that:

- Lead to a deviation from crew or organizational or expectations;
- Reduce safety margins; and
- Increase the probability of adverse operational events on the ground or during flight.

Errors can be the result of momentary diversion of attention (**slip**), or memory failure (**lapse**) induced by an expected or unexpected threat. There are also more deliberate, ***intentional non-compliance*** **errors.** These are often shortcuts used to increase operational efficiency, but in violation of standard operating procedures.

Slips and **lapses** are failures in the execution of an intended action. Slips are actions that do not go as planned, while lapses are memory failures. For example, selecting the probe window heat instead of the APU Bleed is a slip. Forgetting to select the APU Bleed itself, is a lapse. In short both of these would be Unintended errors. **Mistakes** are failures in the plan of action; even if execution of the plan was correct, it would not have been possible to achieve the intended outcome.



Types of Errors

The TEM model classifies errors based upon the primary interaction of the pilot or flight crew at the moment the error is committed. Thus, in order to be classified as:

1) **Aircraft handling error:** the pilot or flight crew must be interacting with the aircraft (e.g. through its controls, automation or systems).
2) **Procedural error:** the pilot or flight crew must be interacting with a procedure (e.g. checklists; SOPs; etc.)
3) **Communication error:** the pilot or flight crew must be interacting with people (ATC; groundcrew; other crewmembers, etc.)

Aircraft handling errors	Examples
Flight control	Incorrect flaps or power settings
Ground navigation	Attempting to turn down wrong taxiway/runway, missed taxiway/runway/ gate, failure to hold short
Manual flying	Hand flying vertical, lateral, or speed deviations
Systems/ Radio/ Instruments	Incorrect GPS, altimeter, fuel switch, transponder or radio frequency settings

Procedural errors	Examples
Briefings	Missed items in the brief, omitted departure, take-off, approach or handover briefing
Callouts	Omitted take-off, descent or approach callouts
Checklist	Performed checklist from memory or omitted checklist, missed items, performed late or at wrong time
Documentation	Wrong weight and balance, fuel information, ATIS or clearance recorded, misinterpreted items on paperwork
Other procedural	Other deviations from regulations, flight manual requirements or standard operating procedures

Communication errors	Examples
Pilot to external	Missed calls, misinterpretation of instructions or incorrect read-backs to ATC, wrong clearance, taxiway, gate or runway communicated
Pilot to pilot	Internal crew miscommunication or misinterpretation

Error Response:

The aim of error management is to have pilots detect errors and respond quickly to them, so that the errors become operationally inconsequential and the risk to safety is minimized.

There are three responses by the crew to manage the errors:

1) *:* An active flight crew response in which an error is detected and managed to an inconsequential outcome;

2) ***Exacerbate:*** A flight crew response in which an error is detected but the crew action or inaction allows it to induce an additional error, Undesired Aircraft State, incident or accident; and

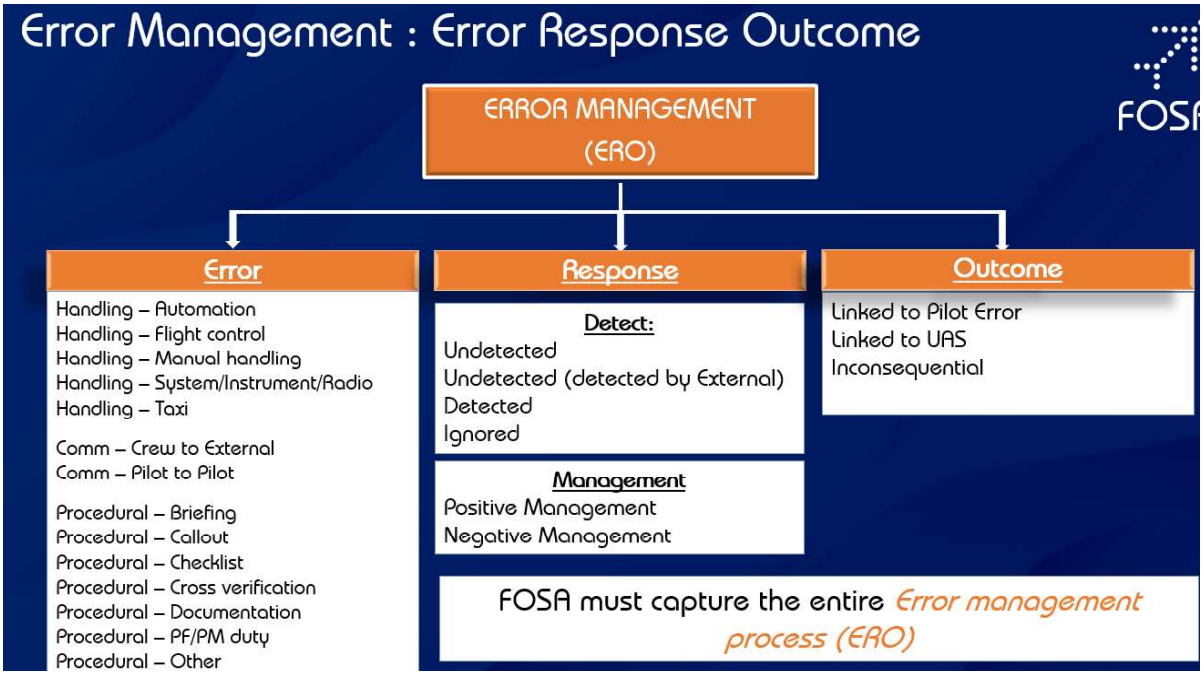
3) ***Fail to respond****:* The lack of a flight crew response to an error because it was either ignored or undetected.

A mismanaged error is one which is linked to or induces an additional error or undesired aircraft state.

For purpose of FOSA, we classify Error Response as

- ***DETECTED (Detected and Acted Upon)***: An active flight crew response in which an error is detected and managed to an inconsequential outcome.
- ***IGNORED (Detected and Not Acted upon)***: The lack of a flight crew response to an error because it was detected and then ignored.
- ***UNDETECTED***: The lack of a flight crew response to an error because it was undetected.
- ***UNDETECTED (Detected by External)***: The lack of or flight crews response to an error because it was detected by an external agency (ATC, Ground Crew etc.)

Error Management:



UAS MANAGEMENT:

Definition:

Undesired aircraft states (UAS) are pilot-induced aircraft position or speed deviations, misapplications of fight controls, or incorrect systems configurations associated with a reduced margin of safety.

For safe fight we must quickly recognise and recover from an undesired aircraft state before it leads to a loss of control or uncontrolled fight into terrain.

Examples of errors and associated undesired aircraft states in operations include:

- mismanagement of aircraft systems (error), resulting in aircraft anti-ice not turned-on during icing conditions (state)
- inappropriate scan of aircraft instruments (error), resulting in an unusual aircraft attitude (state)
- flying a final approach below appropriate threshold speed (error), resulting in excessive deviations from specified performance (state).

Aircraft handling	Examples
Aircraft handling	Vertical, lateral or speed deviations Unnecessary weather penetration Unstable approach Long, floated, firm or of-centreline landings

Ground navigation	Examples
Ground navigation	Runway/taxiway incursions Wrong taxiway, ramp, gate or hold spot Taxi above speed limit

Incorrect aircraft configuration	Examples
Incorrect aircraft configuration	Automation, Engine, Flight control, Systems or weight/ balance events

Management of UAS:

There are three possible crew responses to Undesired Aircraft States:

1) ***Mitigate:*** An active flight crew response to an Undesired Aircraft State that results in the alleviation of risk by returning from the Undesired Aircraft State to safe flight;

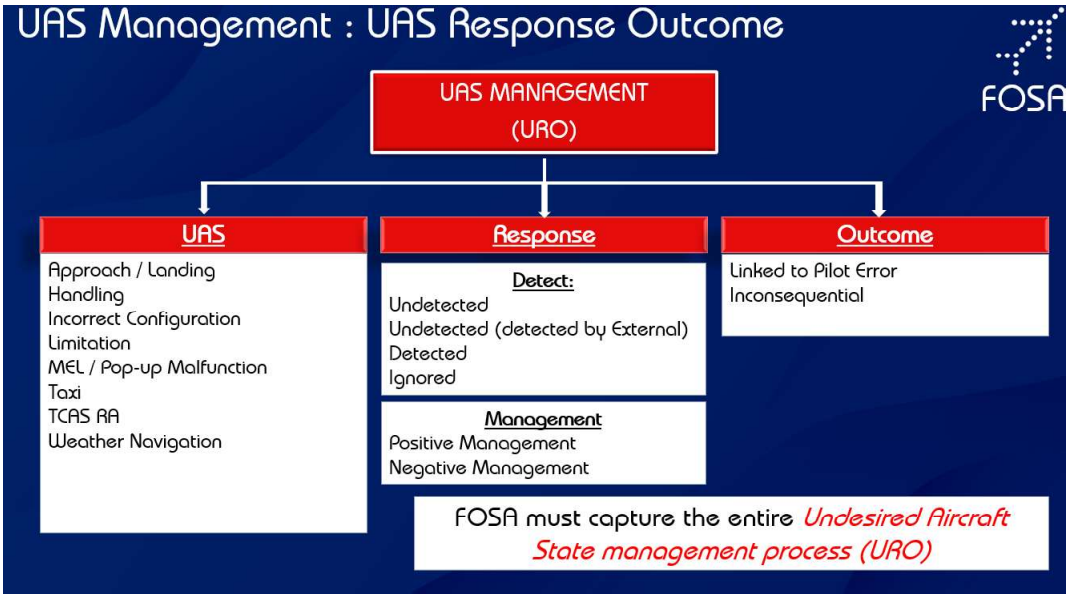
2) ***Exacerbate:*** A flight crew response in which an Undesired Aircraft State is detected, but the flight crew action or inaction allows it to induce an additional error, incident or accident; and

3) ***Fail to respond:*** The lack of an active flight crew response to an Undesired Aircraft State because it was ignored or undetected.

For the purpose of FOSA, we classify UAS management as:

- ***DETECTED (Detected and Acted Upon)*:** An active flight crew response in which an UAS is detected and managed to an inconsequential outcome.
- ***IGNORED (Detected and Not Acted upon)***: The lack of a flight crew response to an UAS because it was detected and then ignored.
- ***UNDETECTED***: The lack of a flight crew response to an UAS because it was undetected.
- ***UNDETECTED (Detected by External)***: The lack of or flight crews response to an UAS because it was detected by an external agency (ATC, Ground Crew etc.)

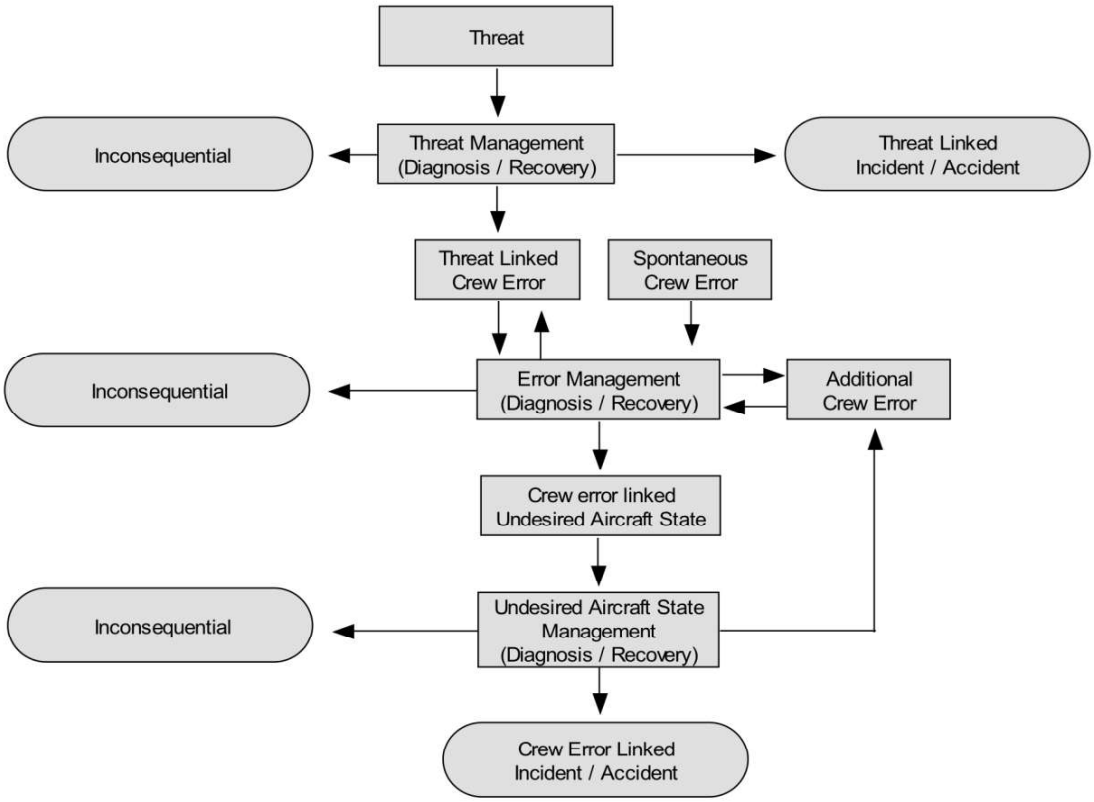
Summary of UAS Management:



CUES to Identify a UAS:

- Aircraft deviations that crews have to detect and act quickly to maintain safety
- Typically involves a length of time that passes without an error being detected
- Crews typically detect an undesired state before the error or errors that contributed to the state have been detected
- Most undesired states are Aircraft Handling Errors followed by Procedural Errors

TEM SUMMARY



COMPETENCY:

Flight crews, as part of their normal operational duties, employ ***countermeasures*** to keep threats, errors and undesired aircraft states from reducing margins of safety in flight operations. Examples of countermeasures would include checklists, briefings, call-outs and SOPs, as well as personal strategies and tactics.

These are personal strategies and tactics, individual and team countermeasures, that typically include canvassed skills, knowledge and attitudes developed by human performance training, most notably, by Crew Resource Management (CRM) training. There are basically three categories of individual and team countermeasures:

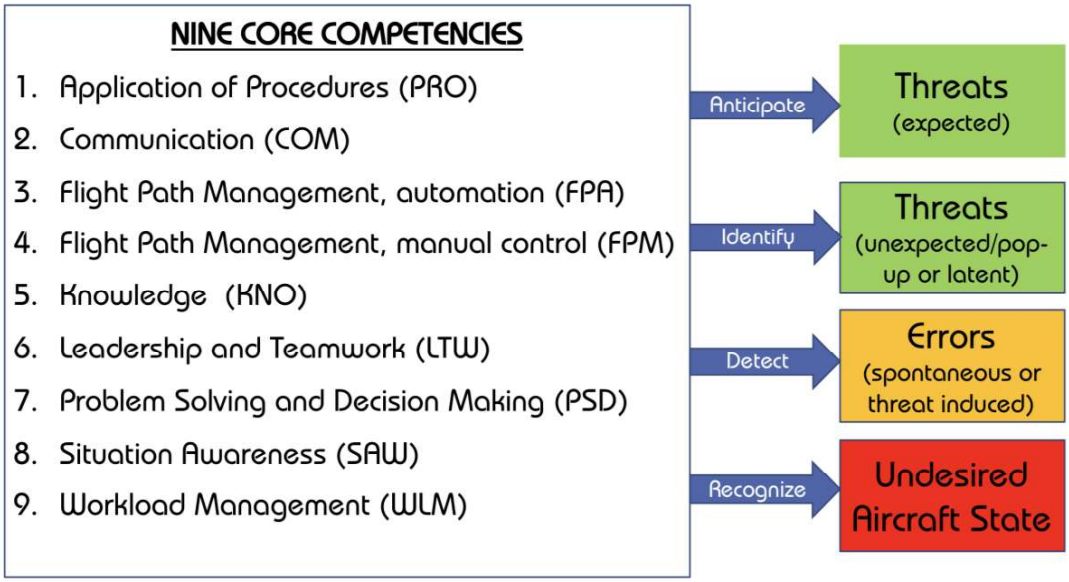
- Planning countermeasures: essential for managing anticipated and unexpected threats;
- Execution countermeasures: essential for error detection and error response;
- Review countermeasures: essential for managing the changing conditions of a flight.

The "ever present rain of threats and errors" challenges flight crews to continuously deploy these countermeasures in order to maintain sufficient safety margins during all phases of flight.

Competencies: State of the Art TEM Countermeasures:

During the last decades industry and regulators have made significant progress in identifying the essences of professional flight crew "work". From the broad term "airmanship", via more advanced systems like "NOTECHS", the worldwide collaborative effort has evolved to the concept of pilots competencies.

ICAO stipulates an agreed set of 9 pilot "Core Competencies", which can be used by operators to describe and measure pilot performance ensuring a harmonized measuring criteria and would enable global evaluation and innovation in safety management.

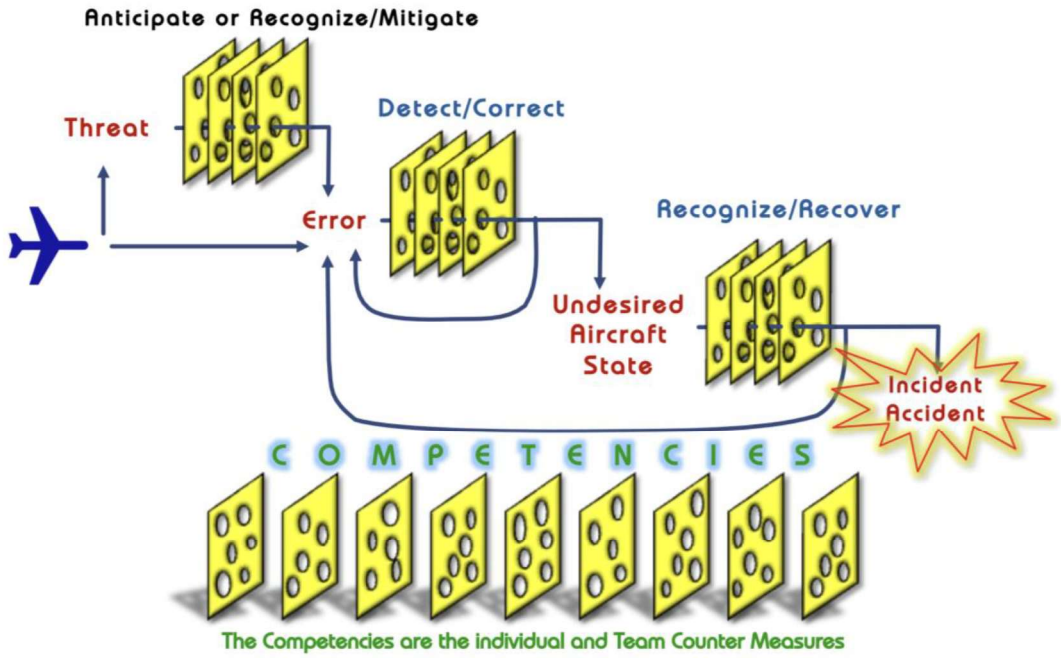


For a modern TEM model, the core competencies constitute today's state-of-the-art enhanced countermeasures.

These are state-of-the-art set of countermeasures, collaboratively developed by an international working group comprised of civil aviation authorities, academic institutions, aircraft original equipment manufacturers, airlines, international organizations, pilot representative bodies and training organizations. Each core competency is well defined by a description and by behavioural indicators. (*Refer to* Annexure III *for the list of competencies along with the observable behaviours.)*

Successful TEM is, therefore, directly related to the competencies as countermeasures.

Flight crew, therefore use these countermeasures (competencies) in managing Threats, Errors or UAS'). The '**response'** and '**outcome'** ("RO") for each Threat, Error and UAS coded is either **managed well (using competencies)** or **mismanaged (where the competencies could have been used)**. For every T/E/UAS that was managed well (**positive management**), observers are required to code the most appropriate competency based on 'observable behaviour' that most accurately depicts the management behaviours observed. Similarly, for every T/E/UAS that was mismanaged (**negative management**), observers are required to code the most appropriate competency based on the 'observable behaviour', if used, would result in a positive outcome. While there could be a combination of competencies uses, observers are required to code the 'most appropriate' one.



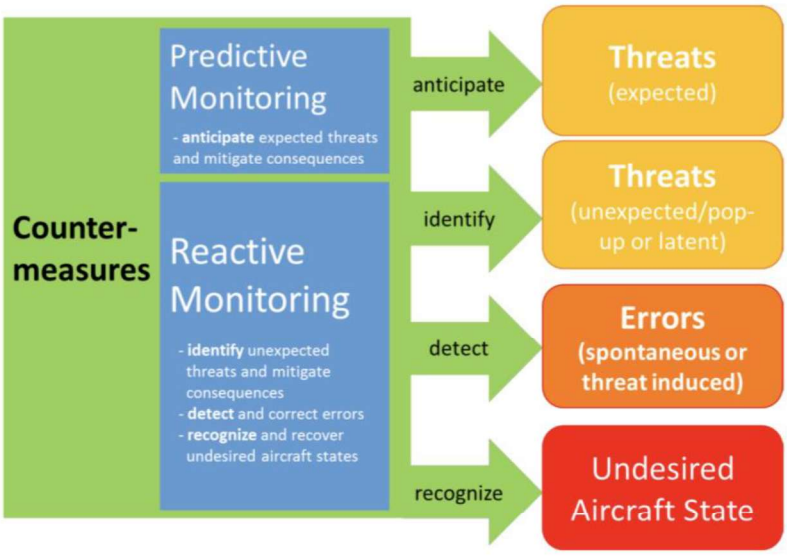
"Monitoring" as a Countermeasure:

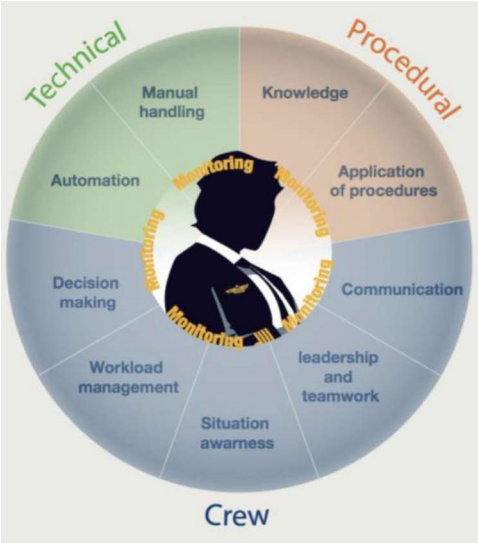
Recognition and management of threats, errors and undesired aircraft states would not be possible without monitoring. Monitoring is, therefore, a key enabler for TEM; it is inherent to countermeasures. Monitoring enables flight crews to manage threats, errors and undesired aircraft states. Monitoring stimulates a continuous interaction (verbal or non-verbal) between flight crew members and between the flight crew and the environment. This dialogue facilitates the application of TEM countermeasures which eventually will enhance TEM performance.

Simply put, Threat and Error Management is not possible without monitoring.

Before the introduction of the ICAO Core Competencies, 'Monitor and Crosscheck' was an important 'countermeasure' under the LOSA TEM framework. In the ICAO set of competencies, we see that monitoring is not a separate (additional) competency. It transcends the competencies and appears in nearly all of them (except KNO),as part of their observable behaviours and is embedded in each of the nine ICAO competencies.

*(Refer competency card : All observable behaviours highlighted and in **bold** are related to monitoring) – Refer to Annexure III for the competency card.*





Monitoring resides in the TEM countermeasures (competencies)

Monitoring is at the center of the 9 core competencies

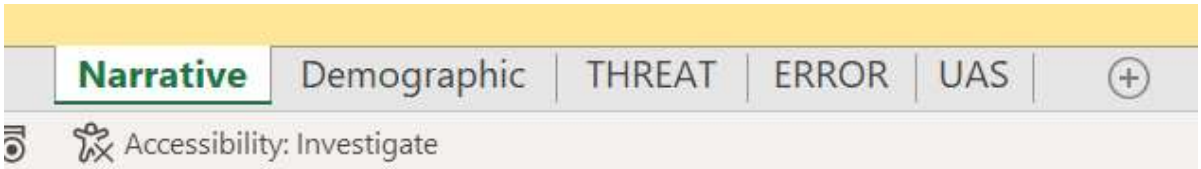
INDIGO FOSA EXCEL WORKBOOK:

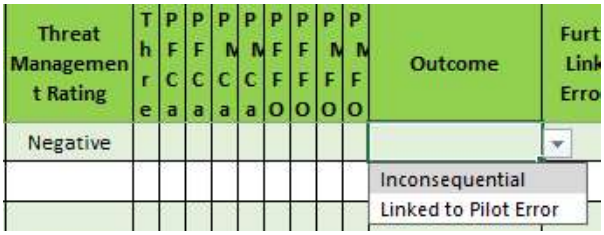
The INDIGO FOSA excel workbook is divided into 5 parts (tabs/sheets):

- ***Narrative***: Describing what the crew did well, what they did poorly and how they managed
- Flight & Crew ***Demographic*** Worksheet: DEPARUTRE/DESTINATION, TYPE, DEP/ARR Time, PIC/FO experience crew familiarity etc
- threats/error/UAS for each phase of the flight.
- ***Threat*** Management Worksheet.
- ***Error*** Management Worksheet.
- ***UAS*** Management Worksheet.

The FOSA Excel File is protected to safeguard and maintain its predictable functioning. At times Unprotecting individual worksheets may be required to edit certain enteries (description/s) in the worksheet

UNPROTECTING THE WORKSHEETS MAY CAUSE THEM TO FUNTION IN AN UNPREDICTABLE MANNER WHICH REMAINS BEYOND THE SCOPE OF THIS GUIDE





- **<u>ERROR MANAGEMENT WORKSHEET</u>:**

Considerations prior to coding an even as an Error:

In order to mark an event as an error it must meet the following criterion:

- Errors may be defined in terms of non-compliance with regulations, Standard Operating Procedures (SOPs) and policies, or unexpected deviation from crew, company or ATC expectations.

- Errors observed may be minor (selecting the wrong altitude into the mode control panel (MCP), but correcting it quickly) or major (forgetting to do an essential checklist).

- Operators also have **policies** that are less proscriptive than **procedures,** where preferred modes of operation are described. Pilots may violate policies without violating SOPs or increasing risk, and under LOSA, this is not defined as an error. However, if the observer feels that violating a policy unnecessarily increases risk to flight safety, it would be defined as an error.

- There are also many decision points on a normal flight that are not defined by SOPs or procedures. However, any time the crew makes a decision that unnecessarily increases risk to flight safety, it is defined as a crew error.

To ensure the Stability and smooth functioning of the Excel Workbook, the *Error Management Worksheet* is limited to 30 rows. If for a particular sector, the errors need to be increased beyond that, please get in touch with the back-end team.

Observers must prioritize coding errors based on the following suggested order of importance -

- Errors linked to threats
- Errors which may be precursors to significant safety events. E.g.- altitude bust, runway incursions, taxiway errors, cabin crew injury rates
- Errors due to the introduction of New Policies & Procedures (things that can't be captured via other programs)
- Any errors which may not be defined by SOPs or procedures but involve an unnecessary increase of risk to flight safety and errors which originate from this.
- Errors originating from one single flow may be coded as one entry.

- **Error ID & Unique Error ID:**

The Error ID (column A) will be pre-filled and user will not be able to make any changes to it. It will remain locked.

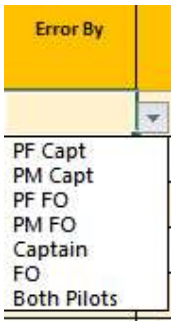
- **Error Description:**

Describe the specific error from the narrative. Observer must remember the description must include a context to help the validating team to comprehend the error.

Example: "During cruise at FL320, the Captain asked the FO to try and obtain FL350 via the HF radio link. A clearance was obtained to climb to FL340. The Captain incorrectly selected FL350 on the FCU." is a detailed description of the error made instead describing it merely as "The Captain selected an incorrect altitude".

- **Error By:**

The observer must select the most appropriate option from the dropdown list indicating by whom the Error was made.



- **Error Type, Error Category & Error Sub Category**

Errors Type (Auto Fill)
Classified as Unintentional & Intentional Non-Compliance

-*Unintentional* = Honest slips / lapses / mistakes

-*Intentional Non-compliance* =

To be coded as intentional Non-Compliance, the event must meet one of the following four conditions:

1. Pilots discussed they were going to break away from SOP
2. Procedural shortcut performed to save time or effort
3. Piloting risk when more conservative options or time was available.
4. Error that happens multiple times within one phase of flight (log this event as one intentional noncompliance).

Error Category (Auto Fill)
Unintentional & Intentional Non-Compliance further Categorised as

-*Unintentional*
HANDLING
COMMunication
PROCEDURAL

-*Intentional Non-compliance*

HANDLING_Int
COMMunication_Int
PROCEDURAL_Int

Error Sub-Category :
Error Category is further sub divided into *Error Sub-Categories*

Error Sub-Category *(Observer Input Required)*

Handling – Automation Flight Control Manual Handling System/ Instrument / Radio Taxi	Handling – Automation_Int Flight Control_Int Manual Handling_Int System_Instrument_Radio_Int Taxi_Int
Comm – Crew to External Pilot to Pilot	Comm – Crew to External_Int Pilot to Pilot_Int
Procedural - Briefing Callout Checklist Cross Verification Documentation PF/PM Duty Other	Procedural - Briefing_Int Callout_Int Checklist_Int Cross Verification_Int Documentation_Int PF/PM Duty_Int Other_Int

Refer to **Types of Error** for definitions.

The ***Error Type*** and ***Error Category*** Column are auto-filled based upon the selection in the ***Error Sub Category*** *Column*. ***Error Type*** and ***Error Category*** Column do not require any selection and are locked for user input.

- **Error Code**

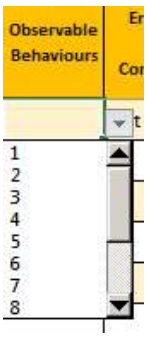
The observer must select the relevant ***Error Code***. Based upon the selection of the ***Error Sub-Category*** column corresponding codes are available to be user selected from the drop-down list.

Refer to Annexure III TEM Code Book for the list of the codes.

- **Error Outcome Competency & associated Observable behaviour:**

The observer must select, based on observation, which observable behaviour most accurately depicts the outcome of the error.

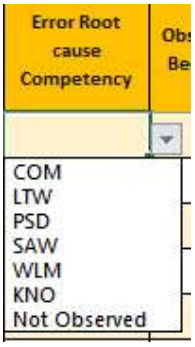
Refer to Annexure III Competency Card
Refer to IPO Model

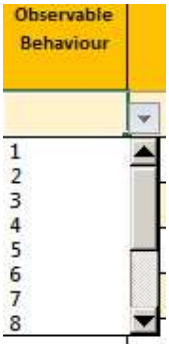


Error Root Cause Competency & associated Observable Behaviour:

The observer must select, based on observation, which observable behaviour most accurately depicts the root cause of the error.

Refer to Annexure III Competency Card
Refer to IPO Model.





Note – Error root cause Competency may not always be observable, in that case *Not Observed* should be selected. A *Not Observed* selection will auto-fill *Observable behavior* column with indicating a further user selection is not required.

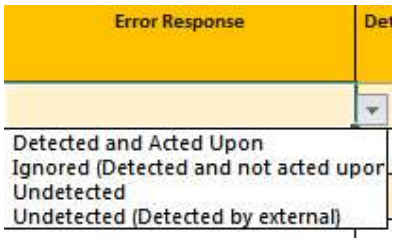
Error Root cause Competency	Observable Behaviour
Not Observed	

If the codes for any reason are different for each crew/ root cause is different, then please code it as a separate error. For Example: While preparing the MCDU the PF Capt forgot to modify the route reserve and the same was not crosschecked by the PM FO. In this case the PF Capt's error would be coded as procedural other error, while the PM FO's error would be coded as a Procedural Cross verification Error.

- **Error Response:**

For the purpose of the FOSA ERROR MANAGEMENT WORKSHEET Error Response have been classified as:

- ***DETECTED (Detected and Acted Upon)***: An active flight crew response in which an error is detected and managed to an inconsequential outcome.
- ***IGNORED (Detected and Not Acted upon)***: The lack of a flight crew response to an error because it was detected and then ignored.
- ***UNDETECTED***: The lack of a flight crew response to an error because it was undetected.
- ***UNDETECTED (Detected by External)***: The lack of or flight crews response to an error because it was detected by an external agency (ATC, Ground Crew etc.)



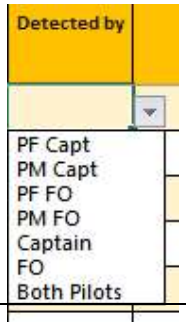
Note: For an error to be coded IGNORED, the observer must have valid observations that confirm that the Flight Crew have decided to ignore an error. The same should be recorded in the error management description.

Error Response	Detected by	Error Management Description	Error Management Rating	Error Management by	PF Capt Competency	PF Capt Observable behaviours	PM Capt Competency	PM Capt Observable Behaviours	PF FO Competency	PF FO Observable Behaviours	PM FO Competency	PM FO Observable Behaviours
Undetected												
Undetected (Detected by external)												

The selection of Undetected or Undetected (Detected by External) will cause further columns not requiring entries to auto-fill with .

- **Detected By:**

The observer must select from the dropdown list the most appropriate option indicating by whom the error was "*Detected by*"



- **Error Management Description:**

A detailed description of how the flight crew managed (or mismanaged) the error must be entered. It must be ensured not to use blanket statements, like, *"The crew managed the error well"*. There should be a detailed explanation as to HOW the crew managed the Error.

Example: "The FO caught the error on his cross check. To make sure of the correct clearance the FO asked ATC to repeat the clearance and the Captain corrected the incorrect setting." is a detailed description of how the crew managed the error of the incorrect altitude setting instead describing it merely as "The Captain selected an incorrect altitude and the FO managed the error by resetting the altitude".

- **Error Management Rating:**

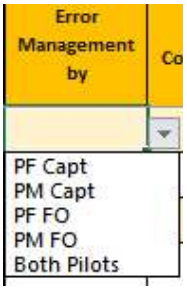
The Observer must select from the Drop-down list ***Positive*** or ***Negative***

The observer must not assign a ***Positive*** error management rating to any error which is linked to another Error or a UAS. Upon Selecting ***Positive*** in ***Error Management Rating*** column, ***Further linked Error ID*** & ***Further linked UAS ID*** column will be auto filled with and be displayed as . Indicating that no input is expected in the column.

Error Managemen t Rating	Erro r Man age men	PF Capt Com pet enc	PF Capt Obs erva ble	PM Capt Com pet enc	PM Capt Obs erva ble	PF FO Com pet enc	PF FO Obs erva ble	PM FO Com pet enc	PM FO Obs erva ble	Erro r Cou tco me	Further linked Error ID	Further linked UAS ID
Positive												

- **Error Management By :**

The observer must select the most appropriate option from the dropdown list.



Based upon the selection in the ***Threat Management By*** Column Respective Columns in PF/PM Competency & Observable behaviors will be Highlighted yellow to prevent any Data entry Error

Error Management by	PF Capt Competency	PF Capt Observable behaviours	PM Capt Competency	PM Capt Observable Behaviours	PF FO Competency	PF FO Observable Behaviours	PM FO Competency	PM FO Observable Behaviours
PF Capt					LTW			
PM Capt								
PF FO								
PM FO								
Both Pilots								

Any selections made in the yellow cells will be shown in RED font indicating an entry was not expected and must be deleted.

NOTE: In the case that the observe selects *Both Pilots* the observer must ensure that the Competency & Root Cause observed are exactly same for both the pilots.

Example: During cruise both pilots were interacting with the observer and due to which they missed carrying out the RVSM Checks. In this case the Error would be coded as: Both Pilots Procedural, etc with a root cause of WLM for both.

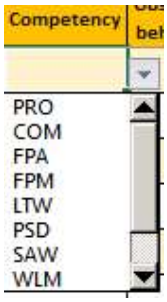
Error Outcome Competency	Observable Behaviours	Error Root cause Competency	Observable Behaviour
PRO		WLM	

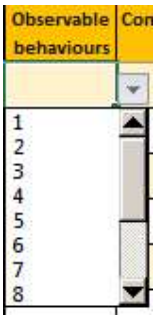
Error Management by
Both Pilots

Error By	Error Type	Error Category	Error Sub Category	Error Code	Error Outcome Competency	Observable Behaviours	Error Root cause Competency
PF Capt	Unintentional	Procedural	Other		PRO		SAW
PM FO	Unintentional	Procedural	CrossVerification		PRO		WLM

- **PF PM Competency and Observable Behaviour:**

Based upon observation, which ***Competency*** & ***Observable behaviour*** most accurately depicts the management (or mismanagement) must be selected from the drop-down list.
Refer to Annexure III Competency Card





- **Error Outcome:**

The possible outcomes from Error management would be:

- Inconsequential (Inconsequential*)*
- Additional Error (*Linked to Pilot Error)*
- Additional UAS (*Linked to UAS)*

Note: A positive Error management can never result in an outcome of *Pilot Error* or *UAS*.
A Negative Error management might not always lead to an additional error or UAS.

Further linked Error ID/UAS ID:

The further linked error/UAS ID to which reference is made should be entered in the relevant cell.
For e.g.- If the coded error leads to another error (Error 4), then enter *E4*. If it led to further two errors (Errors 5 & 6), then enter *E5 E6* in the Further linked error ID column. Similar method may be used to link UAS.

Further linked Error ID	Further linked UAS ID
E4	
E5 E6	
	U1
	U2 U4

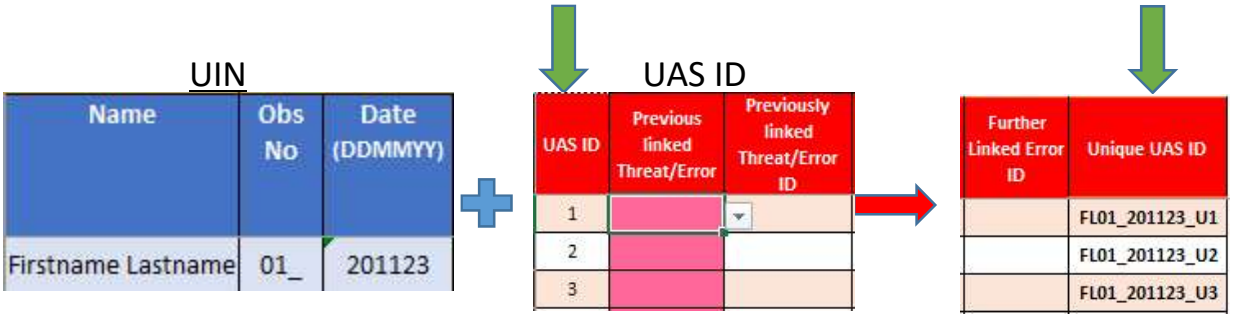
UAS MANAGEMENT WORKSHEET:

UAS ID & Unique UAS ID:

The UAS ID (column A) will be pre-filled and user will not be able to make any changes to it. It will remain locked.

Any New UAS entered or deleted will not cause any change in the "*UAS ID*" column and any other UAS already entered in the sheet.

The sheet will also auto fills a "*Unique UAS ID*" in the last column for every "*UAS ID*".



Note- If, after generating a UAS in the UAS MANAGEMENT WORKSHEET, the observer wants to delete the entry. The observer must select every individual cell, left to right, starting from "PHASE OF FLT" until "FURTHER LINKED UAS ID" and delete any data which the cell/s may contain by selecting the cell and pressing DEL on the keyboard. The workbook is protected and any activities outside the intended work area will generate an Error Message.

- **Previous linked Threat/Error:**

From dropdown list select appropriately the previously linked *Threat/Error* or select *No* if the UAS is not linked to anything.

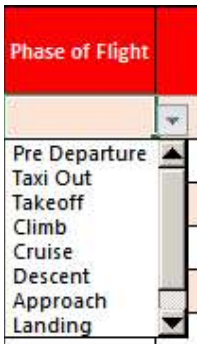
Note: The selection of *NO* will auto-fill *previously linked Threat/Error as* indicating an entry in not required.

- **UAS Description :**

Describe the specific UAS from the narrative. Observer must remember the description must include a context to help the validating team to comprehend the UAS.

- **Phase of Flight:**

Select the phase of flight where the UAS was observed.



- UAS Code:

The UAS classification is straight forward and is based on following categories:

1) Handling (lateral, vertical and longitudinal).
2) Incorrect configuration (Flaps, gear, etc).
3) TCAS RA Handling.
4) Limitations
5) MEL
6) Weather Avoidance.

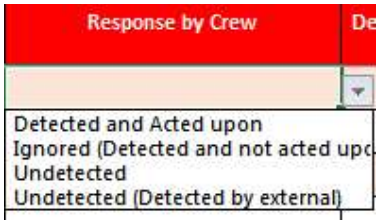
Please refer to the Annexure II TEM Code Book for a detailed list of UAS codes.

To ensure the stability and the smooth functioning of the Excel Workbook, the *UAS MANAGEMENT WORKSHEET* has limited rows. If for a particular sector the observer feels that the number of UAS need to be increased beyond what the WORKSHEET permits, please get in touch with the back-end team.

- Response by Crew:

For the purpose of the FOSA UAS MANAGEMENT WORKSHEET Response by Crew have been classified as:

- ***DETECTED (Detected and Acted Upon)***: An active flight crew response in which an UAS is detected and managed to an inconsequential outcome.
- ***IGNORED (Detected and Not Acted upon)***: The lack of a flight crew response to an UAS because it was detected and then ignored.
- ***UNDETECTED***: The lack of a flight crew response to an UAS because it was undetected.
- ***UNDETECTED (Detected by External)***: The lack of or flight crews response to an UAS because it was detected by an external agency (ATC, Ground Crew etc.)



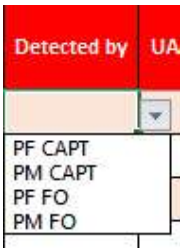
Note: For an error to be coded IGNORED, the observer must have valid observations that confirm that the Flight Crew have decided to ignore an error. The same should be recorded in the error management description.

Response by Crew	Detected by	UAS Management Description	UAS Management	UAS Management by	PF Capt Competency	PF Capt Observable	PM Capt Competency	PM Capt Observable	PF FO Competency	PF FO Observable	PM FO Competency	PM FO Observable	Outcome	Further Linked Error ID
Undetected														
Undetected (Detected by external)														

The selection of Undetected or Undetected (Detected by External) will cause further columns not requiring entries to auto-fill with .

- **Detected By:**

The observer must select from the dropdown list the most appropriate option indicating by whom the UAS was "*Detected by*"

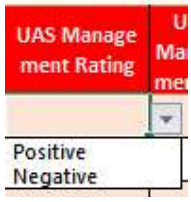


- **UAS Management Description:**

A detailed description of how the flight crew managed (or mismanaged) the UAS must be entered. It must be ensured not to use blanket statements. There should be a detailed explanation as to HOW the crew managed the UAS.

- **UAS Management Rating:**

The Observer must select from the Drop-down list ***Positive*** or ***Negative***

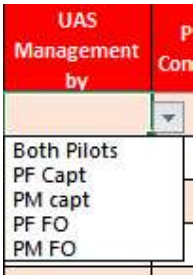


The observer must not assign a ***Positive*** UAS management rating to any UAS which is linked to another Error. Upon Selecting ***Positive*** in **UAS *Management Rating*** column, ***Further linked Error ID*** column will be auto filled with and be displayed as . Indicating that no input is expected in the column.

UAS Manage ment Rating	UAS Mana geme	PF Capt Comp	PF Capt Obse	PM Capt Comp	PM Capt Obse	PF FO Comp	PF FO Obse	PM FO Comp	PM FO Obse	Outc ome	Further Linked Error ID
Positive											

- **UAS Management By:**

The observer must select the most appropriate option from the dropdown list.



Based upon the selection in the ***UAS Management By*** Column Respective Columns in PF/PM Competency & Observable behaviors will be Highlighted yellow to prevent any Data entry Error

UAS Management by	PF Capt Competency	PF Capt Observable behaviours	PM Capt Competency	PM Capt Observable Behaviours	PF FO Competency	PF FO Observable Behaviours	PM FO Competency	PM FO Observable Behaviours
PF Capt					LTW			
PM capt								
PF FO								
PM FO								
Both Pilots								

Any selections made in the yellow cells will be shown in RED font indicating an entry was not expected and must be deleted.

ANNEXURE I:
SAMPLE NARRATIVES

Good Narrative Examples:

Good Narrative Example #1

This narrative provides the desired balance between "what happened" and crew performance. Good opening paragraph providing information about context and crew experience.
The narrative contains landmarks that makes for easier reading and is organized to a timeline. There is also proper performance marker justification and identification.

The CPT arrived 20 mins late due to a transport delay but still gave an unrushed, interactive and comprehensive briefing to the FO and cabin crew. Years of service and experience both with the company and on this particular sector were evident and were used effectively especially as the FO was fairly new on type and had never flown on this sector before.

Preflight Setup
The aircraft was carrying no ADOS, and there were no significant NOTAMS or adverse weather. The ATIS was advising RWY 23 as the active runway, when there was only one runway in PEN, 04/22. This wasn't an issue, only a mild amusement for the crew.
The runway was changed after the FO had set up the FMGC, but well prior to the briefing, and with no time pressure the re-programming was easily accomplished and crosschecked. All ground personnel stood quietly at the door waiting for a break in the crew's activities before speaking to the pilots.

Briefing
The FO gave a good interactive briefing, clearly identifying the terrain as a threat, and briefing the OEI procedure. He also briefed that it was common at this location to have the SID cancelled on line-up, so to be ready for this. The CPT then continued with the "emergency briefing• asking the observer to be part of the crew in the event of an emergency, and not to be afraid to speak up should he be uncomfortable with anything. (Outstanding SOP Briefing / Plans Stated).

Loadsheet
There was a discrepancy between the load sheet PAX numbers. Due to the observer in the jump seat being included in the crew complement of three flight crew and three cabin crew, but also having a boarding pass the CSM was confused with the total number on board in the cabin. The CSM read out to the CPT the wrong figures, but the load sheet figures were correct. The CPT asked the CSM to verify with the supervisors at the gate. The CPT also contacted AMCO services to confirm the PAX and Crew numbers and re-send a copy of the load sheet to the aircraft. The final figures were attained and confirmed to be correct, but inadvertently delayed the push back by 12 minutes. This increased the operational pressure on the crew to speed up the process and keep the On-time-performance.

Before Start / Pushback
The FO called for clearance, but was only given Fl340 and nothing else, no “standby" or other information. The CPT did not wait, but told the FO to request "push & start, “and before the FO could do this the CPT called the ground engineer with Instructions to push. He did not ask if ail doors were secure. The engineer was confused, and the FO stated that he thought the engineer was waiting for the query. At the same time the FO received the push and start clearance, so the CPT called the ground engineer with the correct phrases. (Poor Workload Management)

The aircraft then pushed, and the engines were started with the Alter Start Checklist being read, but not called "completed." This was noticed, and on the very short taxi to crossing the runway, the Before Take-off Checklist was read. The checklist could not be complete, because the crew had not received the clearance, this was "held" but on receiving the clearance, the briefing was not reviewed (Poor Workload Management).

Taxi-Out to Runway

The CPT took the 90' turn onto the parallel taxiway at 17 kts with noticeable lateral "G" force. No comment was made by the FO. (Marginal Runway /Taxiway Management)

The controller held the aircraft at the holding point for four minutes waiting for an arriving aircraft that turned out to be a four place single engine aircraft. Then on line-up, ATC cancelled the SID and gave the crew instructions to maintain runway heading, and climb to 4000 fl. The FO set the new instructions into the FCU, but missed the take-off clearance; the CPT caught this and informed the FO who then asked ATC for confirmation.

Good Narrative Example #2

This narrative provides a good illustration of crew performance intertwined with an event. Again, there is good use of landmarks with slightly different formatting from Exampe#1

******Runway Entry**

The take-off review and before take-off checklist were completed correctly. During the Line-up on runway 19 the PF omitted the standard runway verification call. The PM made the call and no issues resulted from the omission.

*****Take-off Roll**

The aircraft was cleared for take-off on runway 19, with prevailing wind at 210/3 knots. The take-off roll was completed with correct rotation speed and rate; speed on reaching the flight directors was 198 knots, (V2 + 15).

*****Initial Climb to 1000 It. AGL**

Almost immediately after take-off, the captain started a turn up the river, and did not wait until 400 feet, per SOP. About 300 feet, captain reached over to the FCU and selected Flight Path Angle. He did not tell FO of this selection, which is a very unusual mode. In fact, the FO later stated that he has never seen it in his 2 years of being on the Airbus.

Because the Captain did not tell the F/O what he was doing, when the FO noticed this setting, he reselected the more common mode (Heading). Captain then reached over and reselected Flight Path Angle. This led to a brief discussion of whether or not the aircraft would capture NAV in this setting.

*****climb to TOC**

As a result of the mode discussion the FO missed ATC handoff to departure control, a call that was not made until the aircraft levelled off at 5000 feet. Both Captain and FO were heads-down as the aircraft was levelled off correctly by the autopilot. (Marginal Workload Management)

Good Narrative Example #3

This narrative is a good examples of Cruise an Approach briefing conducted prior to TOD: Nice word-picture provided.

***TOC**
All TOC checks-and flight planning were completed correctly, as were both HF frequency and SELCPTL checks.

***En route**
During cruise at FL290 Light and then Moderate Clear Air Turbulence was encountered. The CPT selected the seat belt signs ON. When the turbulence increased to moderate the CPT made a PA requesting the Cabin Crew to be seated as well. After approximately 5 minutes the PF requested climb to FL310. Speed intervention of M0.82 was selected. The turbulence reduced after approximately 15 minutes. Shortly after the aircraft entered smooth flying conditions again the CPT turned OFF the seat belt signs and advised ATC that moderate turbulence was experienced at that level for approximately 15 minutes.

***100 miles to TOD**
The FO made a comprehensive and interactive briefing, with the CPT offering his views of the language threats into BKK, and the possibility of a runway change. The weather was fine with a light breeze down the strip. The FO had 19R in the primary and 19L in the secondary. The missed approach brief was thorough, including the probable holding pattern entry, as well as holding fuel and alternate.

Good Narrative Example #4

Blue Box - This narrative is a good example of a complex Descent/Approach/Land. Again, there is good use of landmarks formatted slightly different from the other examples.

+++TOD - The FO started the descent down lo FL 180 at the FMGC generated TOD point. He used a C.I. of 60 that gave a descent speed of 285 kts / MO.76. ATC quickly cleared the aircraft down to 11000 ft, and at this point the Approach checklist was read. The FO had the CPT query ATC about speed restrictions, and he was informed that there was a limit of 250 kts after SILVA.
+++20000-10000 ft. - the FO used VS of -2000 fpm down to FL145 where he selected 250 kts as the aircraft was approaching SILVA. At FL160, the FO (PF) called ATC to confirm the landing runway. ATC replied 19R, and the CPT did not appear to notice the FO's action. From SILVA the FO used a combination of selected speed and VS, slowing to 230 kts at 10000 ft, as ATC started giving vectors.
+++Transition - Passing 9000 ft., the Transition and Approach checks were completed correctly.
+++Approach • IAF - Al 6500 ft SLATS 15 went out, and at 5000 ft the CPT prompted ATC for further descent. ATC then cleared the aircraft down to 3000 ft with a speed of 190 kts. The aircraft turned base at 4200 ft, and at 2000 ft Flap 15 was selected. ATC asked for 170 kts and informed the crew that there was a preceding aircraft at five miles doing 110 kts. They then gave 160 kts and Flap 20, and when established on the localizer at 10 DME the FO requested the gear. The CPT reminded the FO of the preceding aircraft and at 8.0 DME the aircraft was fully configured at Vapp.

The CPT briefed for a late landing clearance, and the landing checklist was completed at 1500 ft, with ATC landing clearance received at 700 ft. The FO disconnected the autopilot and auto thrust at 700 fl. At 400 ft the aircraft reached 140 kts, or Vapp +13 kts, the CPT called "speed" and at 200 ft the speed was 137 kts. The aircraft continued on speed and profile to land on the 2000 ft markers.

+++Taxi-in - Once again the CPT exceeded 15 kts in the turns, and going through an "S" turn maintained 17 kts with noticeable "G" forces. The FO made no comment about the taxi speed.

+++Park I Offload - Upon arrival at the gate the marshaller, and wing walkers were in position, and no obstructions were observed in the sterile zone. The Shutdown checklist was completed correctly.

Poor Narrative Examples:

Poor Narrative Example #1

Not enough information - If you find yourself wanting to write "routine" or "standard" in a narrative since there were no threats and errors observed, write about 2-3 things that the crew is doing well.

Climb - Routine (Basic SID departure). No threats and errors observed.

Poor Narrative Example #2

Not enough details to be useful - Need to say more about what happened after the FO refused. No word-picture provided. What did the FO say when he refused? the crew is doing well.

3000 ft. - ATC gave the crew a runway change during the approach but the FO refused.

Poor Narrative Example #3

Don't make blanket statements - Always explain why the crew did a good job. Explain why the crew displayed good airmanship.

The crew did a good job during the final approach and landing. Both pilots were good stick and rudder guys. They made a high threat environment "look like walk in the park." I would definitely fly with these guys again. Excellent airman ship!!

Poor Narrative Example #4

Difficult to read - There are no paragraphs or landmarks to help process the story.

CPT arrived 01:15 min before departure, observer arrived around 02:00 hours before ETD. MEL slat sys 2 fault plus a note of missing pages on the Jeppesen txt manual (route manual), crew paid attention to the missing pages, especially the CPT he requested the originals to be barrowed. CFPs were checked quickly and no attention was given to MORAs during the route. WX was forecast TS at the destination (tempo) and TS at the alternate, CPT commented (good weather). Cabin crew reported ready for briefing 75min to ETD. NOTAM check continued with revising drift down procedure for the return Flight mistakenly to the outbound flight. Destination NOTAM was not checked. The rest was just casual paper turning over (the flt documents briefing was done in around 5-8 mins). cabin crew briefing was done in a about 15-20 mins over covered some encouraging points. late crew arrival to the plane (40 min before departure}. there was a time discrepancy in the tech log as engineer found spare 50 hours to the next check (plane was due if not found). was accepted by the CPT without fully attended interest.

Poor Narrative Example #5

Tell a story- Don't tum into a OAR and CVR. If you find yourself doing this, make sure you talk about what the crew did well when they were executing these actions.

Remember, 80% of your narrative should be about TEM and crew performance markers.

Briefing –The captain performed landing briefing at around 18000ft. He briefed by the book thorough.
17000 ft. - The Captain sets ARM VOR, GNS was disengaged.
15500 ft. -ATC: Descend 13000, QNH 2995, Contact 120.45.
15000 ft.- ATC: Descend 5000, leave STE Heading 070.
Transition - 13500ft - CDU message "SEL DES" appeared, the Captain executed that.
13000 It. - The aircraft got under the cloud condition, the FO turned Engine anti-ice off.
10400 It. - The Captain set SPD 250 on MCP, the speed was reduced to 250kt by PMS.
FAF / OM - ATC: Heading 340, Cleared for ILS 32l, Contact TWR. ATC: Report Outer Marker. 2700 It. - The Captain Set Flap 15, SPD 160.
2200 It. - The Captain Set Gear down, SPD 155.
1800 ft. - The Captain Set Flap 28, SPD 145.
1600 It. - The Captain Flap 40, SPD Vtg: 133kt.
1400 It. - ATC: RWY 32L Cleared to land, wind 09005kt.
1200 ft. - Landing checklist was completed.

Poor Narrative Example #6

Do not use T, £, or U landmarks. Observers can sometimes log something as a threat, or error that The LOSA Collaborative will later delete. Good TRO descriptions.

T1: Thunder storm and shower rain were observed at KANTO area. Rain echo was passed HND airport. Another echo line was approaching NRT. TAF forecasted it would be moderate shower rain with thunderstorm during departure time.

TRO1: The crew checked their route and echo. They expected sooner they could depart; it would be better. They decided to depart on normal SID, and look for the area they could fly with Radar when they were in air.

Poor Narrative Example #7

This observer has repeated phase of flight information in the Overall narrative. Do not cut and paste from your phase of flight narratives and say it's a summary.
The Overall narrative should be a brief summary the communication environment, Captain Leadership and any other personal observations that is relevant to overall crew performance.

Overall - Flight Summary of Threats and Errors

Preflight -TVM customs required that the crew sign a customs declaration form, countersigned by the Captain, which is only required for crew disembarking at Indian ports. This requirement is fairly recent familiar with this document and only enforced at TVM, which is one of 6 Indian destinations served by MMM. CPT was not familiar with this document and refused to sign it - wholly understandable as crew were in transit and not disembarking in TVM. Ground staff produced an email chain with respect to this requirement. CPT very reasonably pointed out that it was all with Service Delivery (Cabin Crew management) and not Flight Operations.
Take-off – After T/O PF failed to reduce thrust at TRA. Despite 'LVR CLB" flashing on PFD FO maintained TLA at Flex gate until well after acceleration phase. As a result of this action, acceleration phase was entered with Thrust still in 'MAN FLX'. PF requested F1 (T/0 setting was F2) with thrust in TIO setting.
PNF complied with request to retract flaps. PF reduced thrust partway through flap retraction sequence. Captain Leadership and Communication Environment were marginal at best.

Poor Narrative Example #8

Do not write narratives using text or chat language.

Do not write narratives using text or chat language. @ 500' PF 404 n strtd drifting 1 dot hi@ 400'. Dis wz clld by PNF. @ 200' n 2 dots hi, PF trid 2 crrct by increasn vert desc r8 2 800'. Dis resultd n a lng ldg by about 1000' frm T/D point.

Translation: At 500', PF lost concentration and started drifting one dot high at 400'. This was called by the PNF. At 200' and two dots high, PF tried to correct by increasing the vertical decent rate to 800'. This resulted in a long landing by about 1000' from the touch down point.
404 =clueless or lost

Poor Narrative Example #9

Do not write narratives in ALL CAPS.

Do not write narratives in ALL CAPS.
PRELFIGHT/ TAXI - FLIGHT PLANNING WAS CARRIED OUT AT THE SELF-BRIEF AREA OF THE INTERNATIONAL CREW ROOM WHERE THERE WERE TWO OTHER CREWS UNDERTAKING THE SAME TASK. THERE WERE MINOR DISRUPTIONS BETWEEN CREWS THAT CAUSED A SMALL AMOUNT OF DISTRACTION. THE CREW INTERACTED ON A LIMITED BASIS